

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item →

☒

B: Sampling bias

☐

C: Misunderstanding probabilistic outputs

☐

D: Variations in how different demographics perceive, prioritize, and utilize AI explanations

☒ Correct

Correct! While sampling bias is an algorithmic bias, it is not an inherent human bias or cognitive limitation.

2.

Which of the following are potential solutions to human-centered XAI challenges that can hinder how XAI explanations are perceived, interpreted and acted upon in practice? (Check all that apply):

1 / 1 point

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Analogies and causal frameworks that better align with how humans reason

☒ Correct

Correct! Potential solutions that can help overcome these human-centered XAI challenges include developing explanation techniques that use analogies and causal frameworks that better align with how humans intuitively reason.

☒

Personalized and/or adaptive interfaces

☒ Correct

Correct! Personalized and adaptive interfaces tailoring visualizations to individual user profiles and mental models is a promising way to overcome human-centered XAI challenges.

☒

Interactive, interrogative interfaces that allow users to co-construct explanations through dialogue

☒ Correct

Correct! Enabling interactive, interrogative interfaces that allow users to co-construct explanations through a dialogue with the AI system, versus passive consumption of static reports, can enhance engagement and sensemaking

☒

Developing explanation techniques grounded in familiar concepts

☒ Correct

Correct! A potential solution that can help overcome these human-centered XAI challenges is developing explanation techniques grounded in familiar concepts.

3.

Because there is no one-size-fits-all solution to making AI explainable, it is important to consider a number of things when deciding what XAI solution is right for you. These considerations include all of the following EXCEPT:

1 / 1 point

☐

Your Specific Use Case

☐

Domain Constraints

☐

Trade-offs

☒

E: Knowledge about the XAI method

☒ Correct

Correct! You should not be limited by your current knowledge around a particular XAI method. I encourage you to continue the XAI Specialization here on Coursera or augment your learnings from this course by reading papers or textbooks to better understand how to implement different XAI methods.

4.

All of the following are examples of Interpretable ML models except:

1 / 1 point

☒ LIME

☐

Generative Additive Models (GAMs)

☐

Rule-based Models

☒ Correct

Correct! LIME is a post-hoc local explainable method and is not considered an inherently interpretable ML model.

5.

How does Occam's Razor relate to our discussion of XAI?

1 / 1 point

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If you have two AI models that perform similarly on your evaluation criteria, you should prefer the more complex, less interpretable one.

☒

B: If you have two AI models that perform similarly on your evaluation criteria, you should prefer the simpler, more interpretable one.

☐

C: If you have two AI models that perform similarly on your evaluation criteria, you should prefer to combine the models into an ensemble model.

☒ Correct

Correct! Occam's Razor states that if you have two competing ideas to explain the same phenomenon, you should prefer the simpler one.

6.

In which explainability technique do you train a sparse linear model to mimic how the blackbox model predictions change with perturbations to the input instance?

1 / 1 point

☒ LIME

☐

Anchors

☐

Individual Condition Expectation

☒ Correct

Correct! Local Interpretable Model-agnostic Explanations (LIME) trains a sparse linear model to mimic how the blackbox model predictions change with perturbations to the input instance

7.

I have features that are correlated. Which XAI visualization approach should I (probably) use?

1 / 1 point

☐ Partial Dependence Plots (PDP)

☐

Individual Condition Expectation (ICE) Plots

☒

E: Accumulated Local Effects (ALE) Plots

☒ Correct

Correct! ALE plots focus on local changes in the prediction when a feature value changes, unlike PDPs, which average out the effects over the entire feature space. This allows you to use correlated features.

8.

Which XAI approach represents the model's knowledge in terms of archetypal instances or patterns?

1 / 1 point

☐ Concept Activation Vectors

☒

Prototype-based explanations

☐

Network Dissection

☒ Correct

Correct! Prototype-based explanations either cluster instances into prototypes and explain predictions in terms of similarities to learned prototypes or use criticisms methods to identify which prototypes most contradict a given prediction.

9.

If I want a visualization that shows the pixels or regions of the image that most influence a neural network's predictions, which explainable visualization method should I use?

1 / 1 point

☒ Saliency Maps

☐

Feature Visualization

☐

Accumulated Local Effects (ALE) Plots

☒ Correct

Correct! Saliency maps highlight pixels or words in an input instance that most influence a neural net's predictions.

10.

What are challenges for using XAI in generative AI (select all that apply)?

1 / 1 point

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Scalability issues

☒ Correct

Correct! Scalability and efficiency issues arise when applying complex XAI techniques to massive models with trillions of parameters.

☒

Aligning explanations with human intuitions

☒ Correct

Correct! Aligning explanations with human intuitions by grounding them in recognizable concepts remains an open issue, especially for generative vision models dealing with abstract imagery.

☒

"C": Developing robust evaluation frameworks and metrics for assessing the quality and faithfulness of explanations

☒ Correct

Correct! Developing robust evaluation frameworks and metrics for assessing the quality and faithfulness of explanations for generative models is an active area of research.

☒

Addressing ethical challenges like deception, misinformation, and societal biases encoded in these models

☒ Correct

Correct! Addressing ethical challenges like deception, misinformation, and societal biases encoded in these models is a growing priority where XAI can play an important role.